The equations of motion of a rigid top

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Abstract

In this short note, we introduce the equations governing the motion of a rigid top with its bottom clamped to the origin.

1 Problem Statement

We consider the problem of the "laboratory top", which consists of a symmetric rigid body that rotates around one of its axes while being suspended on a fixed point at its lower end (physically, this could be a pin or a similar support).

The goal is to find the time evolution of the motion described by the principal axis of the top, i.e., the axis around which the object is spinning.

2 Theory

We use, as usual, the Lagrangian formalism to study the proposed system. In this case, we deal with a rigid body, whose Lagrangian has the form

$$L = \tfrac{1}{2} I(\boldsymbol{\omega}, \boldsymbol{\omega}) - U = \tfrac{1}{2} I_{ij} \omega^i \omega^j - U,$$

where I is the bilinear form defined by the inertia tensor of the rigid body, which is symmetric by its definition.

Given the symmetry of I, it is possible to find an orthonormal basis of eigenvectors e_1, e_2, e_3 of I, with respect to which the matrix of I is diagonal, with principal moments of inertia I_1, I_2, I_3 on the diagonal (see [LL76] or [Arn13]).

First, we place our reference frame parallel to the principal axes of inertia of the top at an arbitrary initial time, and we assume that the system is rotating around the axis with moment I_3 with a fixed end. To describe the system's rotations from that instant on, we introduce a coordinate description of rotations of this nature.

Suppose we have a rotation mapping the vectors e_1, e_2, e_3 into the vectors E_1, E_2, E_3. We can describe the rotation through the following sequence [Spi10]:

- The positive angle θ of rotation around the axis e_3 that takes vector e_1 to another vector on the plane $e_1 - e_2$, which generates the so-called line of nodes;
- The angle ϕ of rotation around the line of nodes that takes vector e_3 to vector E_3;
- The angle ψ of rotation around E_3 that takes the generator of the line of nodes into E_1.

This parametrization is known as the *Euler angles*.

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Under this parametrization of rotations, if the angular velocity of the top in the fixed coordinate system we choose is

$$\omega = \omega^1 e_1 + \omega^2 e_2 + \omega^3 e_3,$$

then its components satisfy *Euler's geometric equations* [Spi10]:

$$\begin{aligned} \omega_1 &= \dot{\phi} \sin\theta \sin\psi + \dot{\theta} \cos\psi, \\ \omega_2 &= \dot{\phi} \sin\theta \cos\psi - \dot{\theta} \sin\psi, \\ \omega_3 &= \dot{\phi} \cos\theta + \dot{\psi}. \end{aligned} \tag{1}$$

Taking into account these relations, one can then consider the Lagrangian of the system, which under the action of gravity has the form:

$$L = \frac{1}{2} I_i (\omega^i)^2 - mgl \cos\theta, \tag{2}$$

where l is the distance form the bottom end of the top to its centre of mass. All in all, by Equation 1 and Equation 2, we have a function $L = L(\theta, \omega^1, \omega^2, \omega^3) = L(\theta, \dot{\theta}, \dot{\phi}, \dot{\psi})$.

One can take the Euler–Lagrange equations describing the motion of the system:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}^i}\right) - \left(\frac{\partial L}{\partial q^i}\right) = 0,$$

to a system of partial differential equations of order one, actually leading us to the Hamilton formalism. It all begins by the observation that the knetic energy of the system is always defined by a bilinear form on the space of generalised velocities, so when we compute the derivative with repsect to velocities of a Langrangian whose potential energy does not depend on the generalised velocities, we obtain a linear expression that corresponds to "lowering the indices of the velocities", passing to the dual space:

$$p_i = \frac{\partial L}{\partial \dot{q}^i} = M_{ij}\dot{q}^i, \quad \text{for some} \quad M \in \mathrm{Mat}(n, \mathbb{R}),$$

where n is the amount of generalised coordinates. The quantities p_i are called conjugate momenta.

We will not abuse of the patience of the reader by presenting the complete (but actually direct) calculations here, but one can easily verify that, in the case of the laboratory top with generalised velocities $(\dot{q}^i) = (\dot{\theta}, \dot{\phi}, \dot{\psi})$, we have the following expressions for their conjugate momenta:

$$\begin{pmatrix} p_\theta \\ p_\phi \\ p_\psi \end{pmatrix} = \begin{pmatrix} I_1 \sin^2\psi + I_2 \sin^2\psi & (I_1 - I_2)\sin\theta\sin\psi\cos\psi & 0 \\ (I_1 - I_2)\sin\theta\sin\psi\cos\psi & \sin^2\theta(I_1\sin^2\psi + I_2\cos^2\psi) + I_3\cos^2\theta & I_3\cos\theta \\ 0 & I_3\cos\theta & I_3 \end{pmatrix} \begin{pmatrix} \dot{\theta} \\ \dot{\phi} \\ \dot{\psi} \end{pmatrix}. \tag{3}$$

With the conjugate momenta, we obtain a system of partial differential equations of first order as

$$\dot{p}_i = \frac{\partial L}{\partial q^i}(q^i, p_i). \tag{4}$$

In the case of the rigid top, the function on the right-hand side of this equation is easily seen to be:

$$\begin{aligned} \dot{p}_\theta &= I_2\,\dot{\phi}\cos\theta\cos\psi\,\big(\dot{\phi}\sin\theta\cos\psi - \dot{\theta}\sin\psi\big) + I_1\,\dot{\phi}\cos\theta\sin\psi\,\big(\dot{\phi}\sin\theta\sin\psi + \dot{\theta}\cos\psi\big) - I_3\,\dot{\phi}\sin\theta\,\big(\dot{\phi}\cos\theta + \dot{\psi}\big) + mgl\sin\theta, \\ \dot{p}_\phi &= 0, \\ \dot{p}_\psi &= I_1\,\big(\dot{\phi}\sin\theta\sin\psi + \dot{\theta}\cos\psi\big)\big(\dot{\phi}\sin\theta\cos\psi - \dot{\theta}\sin\psi\big) - I_2\,\big(\dot{\phi}\sin\theta\cos\psi - \dot{\theta}\sin\psi\big)\big(\dot{\phi}\sin\theta\sin\psi + \dot{\theta}\cos\psi\big), \end{aligned} \tag{5}$$

After inverting Equation 3, and replacing the expressions for $\dot{\theta}, \dot{\theta}, \dot{\psi}$ in terms of their momenta in Equation 5, we obtain the *Hamilton's equations* of Equation 4. We do this computationally and solve the system using Runge–Kutta methods.

References

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[Spi10] M. Spivak, *Physics for Mathematicians: Mechanics I* (Publish or Perish, 2010).